SUPPLEMENTARY MATERIAL: ROBUST BACKUP CONTROLLER AND BACKUP SET FOR THE UNCERTAIN PLANAR QUADROTOR MODEL

To simplify the notation used in this section, we use $\lambda_{\min}(\boldsymbol{S})$ and $\lambda_{\max}(\boldsymbol{S})$ to denote the smallest and largest eigenvalues of a symmetric matrix $\boldsymbol{S}$, and we define

$$a_m(\boldsymbol{\theta}) \triangleq \frac{1}{m_0} + \delta_m, \quad a_J(\boldsymbol{\theta}) \triangleq \frac{1}{J_0} + \delta_J, \quad \boldsymbol{e}_1 \triangleq \begin{bmatrix} 1 & 0 \end{bmatrix}^\top,$$
$$\boldsymbol{K}_\vartheta \triangleq \begin{bmatrix} k_\vartheta & k_\omega \end{bmatrix}, \ \boldsymbol{b}_\vartheta \triangleq \begin{bmatrix} 0 & 1 \end{bmatrix}^\top, \ \boldsymbol{A}_\vartheta(a) \triangleq \begin{bmatrix} 0 & 1 \\ -ak_\vartheta & -ak_\omega \end{bmatrix},$$
$$\Delta_\vartheta \triangleq \sqrt{\varrho_\vartheta \boldsymbol{e}_1^\top \boldsymbol{P}_\vartheta^{-1} \boldsymbol{e}_1}, \ \Delta_M \triangleq \sqrt{\varrho_\vartheta \boldsymbol{K}_\vartheta \boldsymbol{P}_\vartheta^{-1} \boldsymbol{K}_\vartheta^\top}, \ \bar{\vartheta} \triangleq \vartheta_{\mathrm{r}} + \Delta_\vartheta.$$

Then, parameter bounds in Assumption 1 induce known constants satisfying

$$\delta_g^- \le \delta_g \le \delta_g^+, \quad \delta_\ell^- \le \delta_\ell \le \delta_\ell^+,$$
$$0 < a_J^- \le a_J(\boldsymbol{\theta}) \le a_J^+, \quad 0 < a_m^- \le a_m(\boldsymbol{\theta}) \le a_m^+,$$

for all $\boldsymbol{\theta} \in \Theta$. The proposition below provides sufficient conditions on the design parameters of the backup controller (38) and the backup set (39) for Assumption 2 to hold for the uncertain quadrotor model (36).

Proposition 1. Consider the uncertain quadrotor model (36), the safety function (37) with the safe set $\mathcal{C}_{\mathrm{S}}$, the backup controller (38), and the backup set $\mathcal{C}_{\mathrm{B}}$ in (39). If the controller gains $k_\vartheta > 0$, $k_\omega > 0$ and the design parameters $\boldsymbol{P}_\vartheta = \boldsymbol{P}_\vartheta^\top \succ 0$, $\boldsymbol{Q}_\vartheta = \boldsymbol{Q}_\vartheta^\top \succ 0$, $\varrho_\vartheta > 0$, $\vartheta_{\mathrm{r}} \in (0, \vartheta_{\max})$, $r_x, r_z > 0$, and $r_\vartheta \in (0, \vartheta_{\max}^2)$, with $\bar{\vartheta} < \pi/2$, are chosen such that

$$\begin{aligned}
&\boldsymbol{A}_\vartheta(a_J^-)^\top \boldsymbol{P}_\vartheta + \boldsymbol{P}_\vartheta \boldsymbol{A}_\vartheta(a_J^-) \preceq -\boldsymbol{Q}_\vartheta, \\
&\boldsymbol{A}_\vartheta(a_J^+)^\top \boldsymbol{P}_\vartheta + \boldsymbol{P}_\vartheta \boldsymbol{A}_\vartheta(a_J^+) \preceq -\boldsymbol{Q}_\vartheta, \\
&\Delta_M \le M_{\max}, \\
&\vartheta_{\mathrm{r}} - \Delta_\vartheta \ge 0, \\
&\vartheta_{\mathrm{r}} + \Delta_\vartheta \le \sqrt{\vartheta_{\max}^2 - r_\vartheta}, \\
&\varrho_\vartheta \ge \lambda_{\max}(\boldsymbol{P}_\vartheta) \left(\frac{2 \left\| \boldsymbol{P}_\vartheta \boldsymbol{b}_\vartheta \right\| F_{\max} \max\{|\delta_\ell^-|, |\delta_\ell^+|\}}{\lambda_{\min}(\boldsymbol{Q}_\vartheta)} \right)^2, \\
&a_m^- F_{\max} \cos(\bar{\vartheta}) - g_0 - \delta_g^+ \ge 0, \\
&\mathrm{e}^{-\kappa r_x} + \mathrm{e}^{-\kappa r_z} + \mathrm{e}^{-\kappa r_\vartheta} \le 1,
\end{aligned} \tag{40}$$

then the moment saturation in (38) is inactive on $\mathcal{C}_{\mathrm{B}}$, $\mathcal{C}_{\mathrm{B}} \subseteq \mathcal{C}_{\mathrm{S}}$ holds, and the backup controller (38) renders $\mathcal{C}_{\mathrm{B}}$ robustly forward invariant for (36) for all $\boldsymbol{\theta} \in \Theta$, i.e., Assumption 2 is satisfied.

Proof. We first consider the pitch dynamics. On the set $\mathcal{C}_{\mathrm{B}}$, $h_{\vartheta,\mathrm{b}}(\boldsymbol{x}) \ge 0$ implies $\boldsymbol{y}_\vartheta^\top \boldsymbol{P}_\vartheta \boldsymbol{y}_\vartheta \le \varrho_\vartheta$, and hence

$$|\vartheta - \vartheta_{\mathrm{r}}| \le \Delta_\vartheta, \quad |\boldsymbol{K}_\vartheta \boldsymbol{y}_\vartheta| \le \Delta_M.$$

Thus, by the fourth and fifth conditions in (40), the pitch angle satisfies $0 \le \vartheta \le \bar{\vartheta} \le \sqrt{\vartheta_{\max}^2 - r_\vartheta}$ on $\mathcal{C}_{\mathrm{B}}$. Moreover, the third condition in (40) yields $|\boldsymbol{K}_\vartheta \boldsymbol{y}_\vartheta| \le M_{\max}$, and hence the moment saturation in (38) is inactive on $\mathcal{C}_{\mathrm{B}}$.

We next verify the robust invariance of the pitch dynamics under the backup controller, which satisfy

$$\dot{\boldsymbol{y}}_\vartheta = \boldsymbol{A}_\vartheta(a_J(\boldsymbol{\theta})) \boldsymbol{y}_\vartheta + \boldsymbol{b}_\vartheta \delta_\ell F_{\max}.$$

Since $\boldsymbol{A}_\vartheta(a)$ depends affinely on a, the first two conditions in (40) imply

$$\boldsymbol{A}_\vartheta(a)^\top \boldsymbol{P}_\vartheta + \boldsymbol{P}_\vartheta \boldsymbol{A}_\vartheta(a) \preceq -\boldsymbol{Q}_\vartheta, \quad \forall a \in [a_J^-, a_J^+].$$

Therefore, with $V_\vartheta \triangleq \boldsymbol{y}_\vartheta^\top \boldsymbol{P}_\vartheta \boldsymbol{y}_\vartheta$, we have

$$\begin{aligned}
\dot{V}_\vartheta &= \boldsymbol{y}_\vartheta^\top \left(\boldsymbol{A}_\vartheta(a_J(\boldsymbol{\theta}))^\top \boldsymbol{P}_\vartheta + \boldsymbol{P}_\vartheta \boldsymbol{A}_\vartheta(a_J(\boldsymbol{\theta})) \right) \boldsymbol{y}_\vartheta \\
&\quad + 2 \boldsymbol{y}_\vartheta^\top \boldsymbol{P}_\vartheta \boldsymbol{b}_\vartheta \delta_\ell F_{\max} \\
&\le -\lambda_{\min}(\boldsymbol{Q}_\vartheta) \|\boldsymbol{y}_\vartheta\|^2 + 2 F_{\max} \|\boldsymbol{P}_\vartheta \boldsymbol{b}_\vartheta\| \max\{|\delta_\ell^-|, |\delta_\ell^+|\} \|\boldsymbol{y}_\vartheta\|.
\end{aligned}$$

On the boundary $h_{\vartheta,\mathrm{b}}(\boldsymbol{x}) = 0$, we have $V_\vartheta = \varrho_\vartheta$, and hence $\|\boldsymbol{y}_\vartheta\| \ge \sqrt{\varrho_\vartheta / \lambda_{\max}(\boldsymbol{P}_\vartheta)} > 0$. Therefore, by the sixth condition in (40), $\dot{V}_\vartheta \le 0$ on the boundary of the zero superlevel set of $h_{\vartheta,\mathrm{b}}(\boldsymbol{x})$. Equivalently, $\dot{h}_{\vartheta,\mathrm{b}}(\boldsymbol{x}) \ge 0$ on this boundary, and the pitch component of $\mathcal{C}_{\mathrm{B}}$ is robustly forward invariant.

We now consider the horizontal component of the motion. Since $0 \le \vartheta \le \bar{\vartheta} < \pi/2$ on $\mathcal{C}_{\mathrm{B}}$, the horizontal dynamics of (36) under (38) satisfy

$$\dot{v}_x = a_m(\boldsymbol{\theta}) F_{\max} \sin(\vartheta) - c_x v_x.$$

On the boundary $h_{v_x,\mathrm{b}}(\boldsymbol{x}) = 0$, we have $v_x = 0$, and therefore

$$\dot{h}_{v_x,\mathrm{b}}(\boldsymbol{x}) = a_m(\boldsymbol{\theta}) F_{\max} \sin(\vartheta) \ge 0,$$

where we used $a_m(\boldsymbol{\theta}) > 0$ and $\vartheta \ge 0$. Hence, the v_x component of $\mathcal{C}_{\mathrm{B}}$ is robustly forward invariant. Moreover, on $\mathcal{C}_{\mathrm{B}}$, $v_x \ge 0$, and hence $\dot{h}_{x,\mathrm{b}}(\boldsymbol{x}) = v_x \ge 0$. Therefore, the surface clearance component of $\mathcal{C}_{\mathrm{B}}$ is robustly forward invariant.

We next consider the altitude component of the motion. On the boundary $h_{v_z,\mathrm{b}}(\boldsymbol{x}) = 0$, we have $v_z = 0$, and the vertical dynamics of (36) satisfy

$$\begin{aligned}
\dot{h}_{v_z,\mathrm{b}}(\boldsymbol{x}) = \dot{v}_z &= -g_0 - \delta_g + a_m(\boldsymbol{\theta}) F_{\max} \cos(\vartheta) \\
&\ge -g_0 - \delta_g^+ + a_m^- F_{\max} \cos(\bar{\vartheta}) \ge 0,
\end{aligned}$$

where we used $0 \le \vartheta \le \bar{\vartheta} < \pi/2$ and the seventh condition in (40). Therefore, the v_z component of $\mathcal{C}_{\mathrm{B}}$ is robustly forward invariant. Furthermore, on $\mathcal{C}_{\mathrm{B}}$, $v_z \ge 0$, thus $\dot{h}_{z,\mathrm{b}}(\boldsymbol{x}) = v_z \ge 0$, and the altitude clearance component of $\mathcal{C}_{\mathrm{B}}$ is also robustly forward invariant.

Finally, we show that $\mathcal{C}_{\mathrm{B}} \subseteq \mathcal{C}_{\mathrm{S}}$. Since $h_{x,\mathrm{b}}(\boldsymbol{x}) \ge 0$ and $h_{z,\mathrm{b}}(\boldsymbol{x}) \ge 0$, we have $h_x(\boldsymbol{x}) \ge r_x$ and $h_z(\boldsymbol{x}) \ge r_z$. Moreover, by the fifth condition in (40), $h_{\vartheta,\mathrm{b}}(\boldsymbol{x}) \ge 0$ implies $h_\vartheta(\boldsymbol{x}) = \vartheta_{\max}^2 - \vartheta^2 \ge r_\vartheta$. Therefore, by the last condition in (40), the smooth minimum safety function (37) satisfies $h(\boldsymbol{x}) \ge 0$ for all $\boldsymbol{x} \in \mathcal{C}_{\mathrm{B}}$, i.e., $\mathcal{C}_{\mathrm{B}} \subseteq \mathcal{C}_{\mathrm{S}}$.

Combining the pitch, horizontal, and altitude invariance results with the input admissibility and $\mathcal{C}_{\mathrm{B}} \subseteq \mathcal{C}_{\mathrm{S}}$, the backup controller (38) renders the backup set (39) robustly forward invariant for (36) for all $\boldsymbol{\theta} \in \Theta$. □